OTC-02 | ROTATING EQUIPMENT

Rotating Equipment Engineering Fundamentals

SUGGESTED DURATION: 5 DAYS | IN-COMPANY / LIVE ONLINE

A structured introduction from process requirement to reliable operation. Participants learn how machinery types, duty conditions, package interfaces and maintenance decisions fit together.

WHO SHOULD ATTEND

Early-career engineers, mechanical designers, operations engineers and maintenance personnel moving into machinery engineering.

WHAT PARTICIPANTS WILL BE ABLE TO DO

- Distinguish pump, compressor, fan and positive-displacement machine applications.
- Prepare a basic design basis and identify essential datasheet information.
- Explain key interfaces between the machine, driver, piping and protection systems.

SUGGESTED PROGRAMME OUTLINE

Day 1 | Requirement and machine selection

Process duty and fluid characteristics; pressure, head, flow, power and efficiency; centrifugal versus positive-displacement machines; equipment lifecycle and operating scenarios.

Day 2 | Pump engineering foundations

Pump curves and system curves; duty selection, NPSH and minimum flow; multistage concepts; casing and impeller arrangements; introduction to API 610 applications.

Day 3 | Compressors, fans and drivers

Gas properties and compression principles; centrifugal and reciprocating behaviour; fans and blowers; motors, couplings, speed control and package utilities.

Day 4 | Mechanical integrity and installation

Shafts, bearings, seals and lubrication; alignment and foundations; piping loads and maintainability; inspection, FAT/SAT and installation checks.

Day 5 | Commissioning and reliability

Start-up readiness, operating envelopes and protective functions; common symptoms and troubleshooting; condition monitoring; lifecycle case review and assessment.

OTC-02 | PRACTICAL APPLICATION & DELIVERY

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PRACTICAL WORKSHOPS

Select a machine

Compare equipment options for a water-transfer duty and a gas-service duty; record assumptions and missing information.

Package interface review

Mark mechanical, electrical and instrumentation interfaces on an illustrative package drawing and prepare a readiness checklist.

PARTICIPANT TOOLS AND TAKEAWAYS

Equipment selection worksheet; basic datasheet checklist; package interface register; lifecycle review checklist.

HOW LEARNING IS DELIVERED AND ASSESSED

Facilitated explanations, worked examples, case discussions and small-group exercises. A short baseline check establishes starting knowledge; a closing knowledge check and practical review assess learning. Assessment can be adapted for operators, technicians or experienced engineers.

CUSTOMISATION AND CLIENT INPUTS

Participant experience profile and examples of machinery used at the plant. Calculations use training data and do not constitute a final equipment design.

The agenda is indicative. Final content, duration, batch size, language and delivery arrangements are agreed after a training-needs discussion. Online delivery uses document reviews, calculations, role-play and case exercises; physical demonstrations require separate site arrangements.

FACILITATOR

Prem Prakash Mishra | Mechanical engineer, advisor and corporate trainer. Brings 39 years of career experience across Kuwait Oil Company, IPCL and NTPC, spanning projects, commissioning, maintenance, reliability, asset integrity and industrial audits. Specialist inputs, where needed, are agreed in the proposal.

COMPLETION AND FOLLOW-UP

An Omtech Corporation participation or completion certificate may be provided under the agreed assessment arrangements. This is company training, not an accredited professional qualification or a guarantee of certification. Post-training support is available with scope and duration agreed in the proposal.

REQUEST A CUSTOMISED TRAINING PROPOSAL

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